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April 2026

1 Orientation of Manifolds

Claim: Let X be an n -dimensional manifold and $A \leq \mathbb{R}^n$. Let $f : A \rightarrow X$ be an injective map. Then $f(A^\circ)$ is open in X .

Proof: Let $x \in A^\circ$. Pick $V(fx)$ such that $V \cong \mathbb{R}^n$. Since f is continuous, pick $f(U) \leq V$, where $U(x) \leq A$ is open in $\mathbb{R}^n$. Restrict f to $g : U \rightarrow V$. Then $g : U \rightarrow V$ is an open map, so $f(A^\circ)$ is open.

Claim: Let $f : X \rightarrow Y$ be continuous injective and $A \leq X$. Then $f(A)^\circ \leq f(A^\circ)$.

Proof: Let $y \in f(A)^\circ$, so pick $V(y) \leq f(A)$, V is open in Y . Then $f^{-1}(V) \leq f^{-1}f(A) = A$, where $y = f(x)$ for some $x \in f^{-1}(V)$. $f^{-1}(V)$ is open, so $f(A)^\circ \leq f(A^\circ)$.

Claim: Let $A \leq \mathbb{R}^n$ be compact and $f : A \rightarrow X$ an injective map (also an embedding). Then $f(A)^\circ = f(A^\circ)$.

Proof: We have $f(A^\circ) \leq f(A)^\circ$. The previous gives $f(A)^\circ \leq f(A^\circ)$, so $f(A^\circ) = f(A)^\circ$.

Claim: Let $x_0 \in X$. There exists $U(x_0)$ such that $X - U$ is a SDR of $X - x_0$.

Proof: Pick $f : E^n \rightarrow X$ a homeomorphism with $f(0) = x_0$. Let $U = f((E^n)^\circ) = f(E^n)^\circ$. Let $K : (E^n - 0) \times I \rightarrow E^n - 0$ be the standard SDR of $E^n - 0$ onto S^{n-1} . Define $H : (X - x_0) \times I \rightarrow (X - x_0)$ by

$$H(x, t) = \begin{cases} fH(f^{-1}(x), t) & \text{if } x \in f(E^n), \\ x & \text{if } x \in X - U. \end{cases}$$

Note that $f(E^n), X - U$ are closed and $f(E^n) \cup (X - U) \geq f(E^n) \cup (X - f(E^n)) = X$. Further, $f(E^n) \cap (X - U) = f(E^n) - f((E^n)^\circ) = f(S^{n-1})$, so H is continuous (since points of $S^{n-1} \leq E^n$ are fixed by K).

We have the ladder

$$\begin{array}{ccccccccc}
H_q(X - U) & \longrightarrow & H_q(X) & \longrightarrow & H_q(X, X - U) & \longrightarrow & H_{q-1}(X - U) & \longrightarrow & H_{q-1}(X) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
H_q(X - x_0) & \longrightarrow & H_q(X) & \longrightarrow & H_q(X, X - x_0) & \longrightarrow & H_{q-1}(X - x_0) & \longrightarrow & H_{q-1}(X)
\end{array}$$

The rows are exact and the columns besides the middle are isomorphisms. By the Five lemma, $H_q(X, X - U) \rightarrow H_q(X, X - x_0)$ is an isomorphism.

Note that $(X, X - U)$ is a collared pair, so $H_q(X, X - U) = H_q^\#(X/(X - U)) = H_q^\#(S^n)$. In particular, for $q = n$, we have $H_n(X, X - x_0) = R$, as desired.

Continuation lemma. Given $x_0 \in X$ and $\alpha_{x_0} \in H_n(X, X - x_0)$, there exists $U(x_0)$ and $\alpha \in H_n(X, X - U)$ such that $j_x^U : H_n(X, X - U) \rightarrow H_n(X, X - x_0)$ has $j_x^U(\alpha) = \alpha_{x_0}$.

Proof: Let a denote a relative cycle representing α_{x_0} , so $|\partial a| \leq X - x_0$, so $x_0 \leq X - |\partial a| = U$. Take $\alpha \in H_n(X, X - U)$ to be the homology class of a relative to $X - U$.

Claim: Let $p : E \rightarrow B$ be a covering map. Then p is open.

Proof: Let $U \leq E$ be open, $e \in E$, and $b = p(e)$. Pick $V(b)$ evenly covered by p ; let $\{V_\alpha\}$ be a partition of $p^{-1}V$ into slices. Let $e \in V_{\alpha_0}$. Then $V_{\alpha_0} \cap U$ is mapped homeomorphically to $p(V_{\alpha_0} \cap U)$, which is open in V , and therefore open in B .

Claim: Every connected unorientable manifold has a connected two-fold orientable manifold as a covering space.

Proof: Let X be an unorientable manifold. Let $E = \{(x, \alpha_x)\}$, where $x \in X$ and α_x is a generator of $H_n(X, X - x)$.

Let (U, α_U) be a local R -orientation of X , ie $H_n(X, X - U) \rightarrow H_n(X, X - x)$ is an isomorphism for all $x \in U$. Let $\langle U, \alpha_U \rangle = \{(x, \alpha_x) : x \in U, \alpha_x = j_x^U(\alpha_U)\}$. By the Local Continuation lemma, the sets $\langle U, \alpha_U \rangle$ cover E .

Suppose $(x, \alpha_x) \in \langle U, \alpha_U \rangle \cap \langle U', \alpha_{U'} \rangle$, so $j_x^U(\alpha_U) = j_x^{U'}(\alpha_{U'}) = \alpha_x$. Pick $U''(x) \leq U \cap U'$ such that $H_n(X, X - U'') \rightarrow H_n(X, X - y)$ is an isomorphism for all $y \in U''$; let $\alpha_{U''}$ be the local continuation of α_x . I claim that $j_y^U(\alpha_U) = j_y^{U''}(\alpha_{U''})$. Claim: $j_{U''}^U(\alpha_U) = \alpha_{U''}$. We have $j_x^U(\alpha_U) = j_x^{U'}(\alpha_{U'}) \rightarrow j_x^{U''} j_{U''}^U(\alpha_U) = j_x^{U''} j_{U''}^{U'}(\alpha_{U'}) \rightarrow j_{U''}^U(\alpha_U) = j_{U''}^{U'}(\alpha_{U'})$. $j_x^U(\alpha_U) = j_x^{U''}(\alpha_{U''}) = j_x^{U''} j_{U''}^U(\alpha_U) = j_x^{U''}(\alpha_{U''}) \rightarrow \alpha_{U''} = j_{U''}^U(\alpha_U)$. Then $\langle U'', \alpha_{U''} \rangle \leq \langle U, \alpha_U \rangle \cap \langle U', \alpha_{U'} \rangle$.

$\langle U', \alpha_{U''} \rangle$, so we have a topology on E .

Let $p(x, \alpha_x) = x$. I claim that p is continuous. Let $x \in X$. Pick $U(x)$ that is a local continuation of x and α_x a generator of $H_n(X, X - x)$. p is surjective. Then $p(\langle U, \alpha_U \rangle) = U$, so $p^{-1}U = \langle U, \alpha_U \rangle \cup \langle U, -\alpha_U \rangle$, so p is continuous and has fiber 2.

Claim: p maps $\langle U, \alpha_U \rangle$ homeomorphically to U . p is injective on this. Suppose $\langle V, \alpha_V \rangle \leq \langle U, \alpha_U \rangle$, so we have $\alpha_V = j_V^U(\alpha_U)$.

Pick $(x, \alpha_x) \in \langle V, \alpha_V \rangle$, so $\alpha_x = j_x^V(\alpha_V)$. We have $\alpha_x = j_x^V(\alpha_V) = j_x^U(\alpha_U) = j_x^V j_V^U(\alpha_U) \rightarrow \alpha_V = j_V^U(\alpha_U)$ (for some x in the intersection). $\langle V, \alpha_V \rangle$ is mapped to V , open in X .

Let $p : E \rightarrow B$ be a covering map such that B is locally path connected. Is E locally path connected?

Ans: Let $e \in E$, $b = pe$. Pick V evenly covered by p ; let $\{V_\alpha\}$ be a partition of $p^{-1}V$ into slices. Pick $W(e) \leq V$ open path connected, so by restricting the maps, we have that $\{W_\alpha\}$ is a partition of W into slices. Then W_α is path connected containing e for some α (and open in E since $p^{-1}V$ is open), so E is locally path connected.

E has at most two connected components. If E has two components E_1, E_2 , then we have covering maps $p_1 : E_1 \rightarrow B, p_2 : E_2 \rightarrow B$, each of which is injective.

I claim that E is orientable. Let $(x, \alpha_x) \in E$. Pick (U, α_U) a local continuation of α_x . Consider the covering $\langle U, \alpha_U \rangle$ of E . Pick the associated generator of $\langle U, \alpha_U \rangle$ to be α_U . If $(x, \alpha_x) \in \langle U, \alpha_U \rangle \cap \langle U', \alpha_{U'} \rangle$, then by construction $j_x^U(\alpha_x) = j_x^{U'}(\alpha_{U'})$. Hence, E is orientable.

If E is orientable, so is its components E_1, E_2 . Then $p_1 : E_1 \rightarrow B$ is a homeomorphism, absurd, so E is connected.

Claim: if $\pi_1(X)$ has no subgroup of index 2, then X is orientable.

Define the covering $p : E \rightarrow X$ as before. Suppose FTSOC E is connected. Then $\pi_1(E) \leq \pi_1(X)$ is a subgroup of index 2, absurd.

Hence, E has two components, both of which are orientable. Then $p_1 : E_1 \rightarrow B$ is a homeomorphism between an orientable space and X , so X is orientable.

Let $X^0 = \{(x, \alpha_x) : x \in X, \alpha_x \in H_n(X, X - x_0)\}$, so α_x need not be a generator. Let $U \leq X$ be such that $H_n(X, X - U) \rightarrow H_n(X, X - x)$ is an isomorphism. Let $\alpha_U \in H_n(X, X - U)$ and define $\langle U, \alpha_U \rangle = \{(x, \alpha_x) : x \in U, \alpha_x = j_x^U(\alpha_U)\}$. Then I claim the sets $\langle U, \alpha_U \rangle$ form a basis of X^0 .

Proof: Suppose $\langle x, \alpha_x \rangle \in X^0$. Pick $U(x)$ such that $H_n(X, X-U) \rightarrow H_n(X, X-y)$ is an isomorphism for all $y \in U$.

Let $X = \{x_0\}$. $H_0(X) = \mathbb{Z}$. $X^0 = \{(x_0, z) : z \in \mathbb{Z}\}$. $H_0(X, X-x_0) = H_0(X, \emptyset) = H_0(X) = \mathbb{Z}$. Every point is open, so $X = \mathbb{Z}$ with the discrete topology.

Let X^0 be the orientation sheaf of X . Let $\mathbb{Z}_{\geq 0}$ have the discrete topology. The map $v : X^0 \rightarrow \mathbb{Z}_{\geq 0}$ is continuous. Suppose $s_1, s_2 : A \rightarrow X^0$ are sections. Then

$$\begin{array}{ccccc} A & \longrightarrow & X^0 & \longrightarrow & \mathbb{Z}_{\geq 0} \\ & & & \searrow & \\ & & & & \mathbb{Z}_{\geq 0} \longrightarrow X \times \mathbb{Z}_{\geq 0} \longrightarrow X^0 \end{array}$$

so $s_1 + s_2$ is indeed a section. Similarly, λs is a section for $\lambda \in \mathbb{Z}$.

Suppose there exists a section $s : X \rightarrow v^{-1}(1)$. For each $x \in X$, pick $x \in \langle U, \alpha_U \rangle$ such that $j_x^U(\alpha_U) = s'(x)$. I claim that $\langle U, \alpha_U \rangle$ is an R -orientation system of X . Note that $s'(x)$ gives the local R -orientation of X . Suppose $x \in U \cap U'$. Since s is continuous, pick $U''(x)$ such that $s(U'') \leq \langle U, \alpha_U \rangle \cap \langle U', \alpha_{U'} \rangle$. Hence, $U'' \leq U \cap U'$ and $s'(x) = j_x^U(\alpha_U) = j_x^{U'}(\alpha_{U'})$.

Claim: There is a canonical homomorphism $j_A : H_n(X, X-A) \rightarrow \Gamma A$ given by $j_A(\alpha)(x) = (x, j_x^A(\alpha))$ for $x \in A$.

We show the map $j_A(\alpha) : A \rightarrow X^0$ is continuous. Let a be a relative cycle representing α . Let $U = X - |\partial a|$, so U is open containing A . Let $\alpha_U \in H_n(X, X-U)$ be the homology class of a relative to $X-U$, so α_U induces α under the inclusion of $X-U$ into $X-A$. Given $x \in A$, consider open neighborhoods V of x such that $V \leq U$ and j_x^A has a unique continuation of α_V over V .

Let $A \leq \mathbb{R}^n$ be compact with k connected components. Then k equals $n-1$ st Betti number of $\mathbb{R}^n - A$ for $n \geq 2$.

$\mathbb{R}^n$ is orientable, so $\mathbb{R}^n$ is orientable along A . Then $H_n(\mathbb{R}^n, \mathbb{R}^n - A) = \mathbb{Z}^k$. We have the exact sequence

$$H_n(\mathbb{R}^n) \rightarrow H_n(\mathbb{R}^n, \mathbb{R}^n - A) \rightarrow H_{n-1}(\mathbb{R}^n - A) \rightarrow H_{n-1}(\mathbb{R}^n)$$

Since $n \geq 2$, $H_n(\mathbb{R}^n) = H_{n-1}(\mathbb{R}^n) = 0$, so $H_{n-1}(\mathbb{R}^n - A) = H_n(\mathbb{R}^n, \mathbb{R}^n - A) = \mathbb{Z}^k$, so $\beta_{n-1} = k$ for $\mathbb{R}^n - A$.

Let X be a compact connected manifold. Assume R has the property that

if $a \in R - 0$ such that $ua = a$ for some $u \in R^\times$, then $u = 1$ (so if R is an integral domain works). Then

$$H_n(X) = \begin{cases} R & \text{if } X \text{ is } R\text{-orientable,} \\ 0 & \text{otherwise.} \end{cases}$$

Suppose X is R -orientable. Then X is R -orientable along itself. X is closed compact. Then $H_n(X) = H_n(X, X - X) = R$, since X has one component.

Conversely, suppose $H_n(X) = R$. Consider the projection $H_n(X) \rightarrow H_n(X, X - x)$, which I claim is an isomorphism. Claim: $H_n(X - x) = 0$ and $H_{n-1}^\#(X - x) = 0$. Suppose $X = S^n$. We have $H_n(S^n - x) = 0$ since $S^n - x \cong \mathbb{R}^n$ ($n \geq 1$). Also $H_{n-1}^\#(\mathbb{R}^n) = 0$.

Let X be a compact connected manifold such that $H_n(X) = R$. Let $x \in X$. Then $H_n(X - x), H_{n-1}^\#(X - x) = 0$.

2 Orientability Revisited

Claim: Let X be an n -dimensional manifold. Then $H_n(X, X - x) = R$.

Proof: Pick $U(x) \cong (E^n)^\circ$. I claim that $H_n(U, U - x) \rightarrow H_n(X, X - x)$ is an Excision. Indeed, $\overline{X - U} = X - U \leq (X - x)^\circ$. We have that $\partial : H_n(U, U - x) \rightarrow H_{n-1}^\#(U - x)$ is an isomorphism since $H_n(U) = H_{n-1}^\#(U) = 0$. $H_{n-1}^\#(U - x) = H_{n-1}^\#(S^{n-1}) = R$, since $U - x$ SDR's onto a homeomorph of S^{n-1} .

3 Cohomology

Transposition need not preserve exactness. Let $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$. Suppose FTSOC that exactness is preserved. We have the sequence $0 \rightarrow 0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$, so $\phi : \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \rightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})$ given by $\phi(a^\#)(x) = a^\#(2x) = 2ax = (2a)^\#(x)$, so $\phi(a^\#) = (2a)^\#$, so ϕ is not surjective, absurd.

Let M be a module, $M_1 \leq M_2 \leq M$. For $N \leq M$ denote by $\text{Ann}(N) = \{\varphi \in M^* : \varphi(N) = 0\}$. Then $(M_2/M_1)^* \cong \text{Ann}(M_1)/\text{Ann}(M_2)$.

Proof: Let $\varphi \in (M_2/M_1)^*$. Denote by $p : M_2 \rightarrow M_2/M_1$ the projection. Define $\Phi : (M_2/M_1)^* \rightarrow \text{Ann}(M_1)/\text{Ann}(M_2)$ by $\Phi(\varphi) = \varphi p$ under the quotient $\text{Ann}(M_1)/\text{Ann}(M_2)$. For any $x \in M_1$, we have $\varphi p(x) = 0$, so $\varphi p \in \text{Ann}(M_1)$. Conversely, if $\varphi' \in \text{Ann}(M_1)$, there exists $\varphi \in (M_2/M_1)^*$ such that $\varphi' = \varphi p$. If $\varphi p \in \text{Ann}(M_2)$, then $\varphi = 0$. Hence, Φ is an isomorphism.

4 Ext

Let $f : M \rightarrow M'$ and C, C' be resolutions of M, M' . Then there exists a chain map $f : C \rightarrow C'$ such that the following diagram commutes

$$\begin{array}{ccc} C_0 & \xrightarrow{f_0} & C'_0 \\ \downarrow \epsilon & & \downarrow \epsilon' \\ M & \xrightarrow{f} & M' \end{array}$$

Proof: We induct on q . Fix a basis $\{x_\alpha\}$ of C_0 . Pick $y_\alpha \in (\epsilon')^{-1}(f\epsilon(x_\alpha))$. Define $f_0(x_\alpha) = y_\alpha$. Then $\epsilon'f_0(x_\alpha) = \epsilon'y_\alpha = f\epsilon(x_\alpha) \rightarrow \epsilon'f_0 = f\epsilon$. Rename $\{x_\alpha\}$ to be a basis of C_1 . Note that $\epsilon'f_0\partial_1(x_\alpha) = f\epsilon\partial_1(x_\alpha) = 0$. Since $\ker \epsilon' = \text{im } \partial'_1$, pick $y_\alpha \in (\partial'_1)^{-1}f_0\partial(x_\alpha)$ and define $f_1(x_\alpha) = y_\alpha$. Then $\partial'_1f_1(x_\alpha) = \partial'_1y_\alpha = f_0\partial(x_\alpha) \rightarrow \partial'_1f_1 = f_0\partial$.

Suppose f_k are defined for $k \leq q$ and $q \geq 1$. Then by the same argument as f_1 , we can pick f_{q+1} such that

$$\begin{array}{ccc} C_{q+1} & \xrightarrow{f_{q+1}} & C'_{q+1} \\ \downarrow \partial_{q+1} & & \downarrow \partial'_{q+1} \\ C_q & \xrightarrow{f_q} & C'_q \end{array}$$

Suppose $f : A \rightarrow B, f' : A' \rightarrow B'$. Then we have the commutative diagram

$$\begin{array}{ccc} (B \oplus B')^* & \longrightarrow & B^* \oplus B'^* \\ \downarrow \{f, f'\}^* & & \downarrow \{f^*, f'^*\} \\ (A \oplus A')^* & \longrightarrow & A^* \oplus A'^* \end{array}$$

Let $g : B \oplus B' \rightarrow C$. The top right composition is $g \mapsto (gi_Bf, gi_{B'}f')$. The bottom left composition is $g \mapsto g\{f, f'\} \mapsto (g\{f, f'\}i_A, g\{f, f'\}i_{A'})$.

Let $f : C \rightarrow C'$ be a chain map. The mapping cone Cf is the complex $(Cf)_q = C'_q \oplus C_{q-1}$. The boundary map is given by $\partial_q^f(x, y) = (\partial'_q x + f_{q-1}y, -\partial_{q-1}y)$. We check that

$$\begin{aligned} \partial_{q-1}^f \partial_q^f(x, y) &= \partial_{q-1}^f(\partial'_q x + f_{q-1}y, -\partial_{q-1}y) \\ &= (\partial'_{q-1}(\partial'_q x + f_{q-1}y) + f_{q-2}(-\partial_{q-1}y), -\partial_{q-2}(-\partial_{q-1}y)) \\ &= (\partial'_{q-1}f_{q-1}y - f_{q-2}\partial_{q-1}y, 0) = (0, 0). \end{aligned}$$

We form the exact sequence of chain complexes $0 \rightarrow C' \xrightarrow{i} Cf \xrightarrow{j} C^+ \rightarrow 0$, where $ix = (x, 0)$, $j(x, y) = y$. We define $C_q^+ = C_{q-1}$ and $\partial_q^+ = -\partial_{q-1}$. Note that $H_q(C^+) = \ker \partial_q^+ / \text{im } \partial_{q+1}^+ = \ker \partial_{q-1} / \text{im } \partial_q = H_{q-1}(C)$. It is easy to check that the above sequence is exact.

Define a connecting homomorphism $\partial : H_q(C^+) \rightarrow H_{q-1}(C')$. Thus, we have the long exact sequence

$$H_q(C') \rightarrow H_q(Cf) \rightarrow H_{q-1}(C) \rightarrow H_{q-1}(C') \rightarrow \dots$$

The map $H_q(C) \rightarrow H_q(C')$ is given by $H_q(f) : H_q(C) \rightarrow H_q(C')$.

Claim: Suppose $H(C) = 0$ and R is a PID. Then $1_C \simeq 0$.

Proof: The sequence of modules $0 \rightarrow Z_q \rightarrow C_q \rightarrow B_{q-1} \rightarrow 0$ is exact. Since C_{q-1} is free, $B_{q-1} \leq C_{q-1}$, and R is a PID, we have B_{q-1} is free and therefore projective. The sequence splits, so write $C_q = Z_q \oplus B_{q-1}$. Since $H_q(C) = 0$, we have $Z_q = B_q$. Rewrite $C_q = Z_q \oplus Z_{q-1}$ and let $k_q : Z_{q-1} \rightarrow C_q$ be from the splitting.

Define $\psi_q : Z_q \oplus Z_{q-1} \rightarrow C_q$ by $\psi_q(x, y) = x + k_q y$. Let $\{Z_q \oplus Z_{q-1}\}$ have the boundary map $\partial(x, y) = (y, 0)$. It is easy to see that $\partial^2 = 0$. I claim that $\{\psi_q\}$ is a chain isomorphism. Indeed, ψ_q is an isomorphism by construction and

$$\begin{array}{ccc} Z_q \oplus Z_{q-1} & \xrightarrow{\psi_q} & C_q \\ \downarrow \partial & & \downarrow \partial \\ Z_{q-1} \oplus Z_{q-2} & \xrightarrow{\psi_{q-1}} & C_{q-1} \end{array}$$

commutes since the corresponding diagram of objects is

$$\begin{array}{ccc} (x, y) & \xrightarrow{\psi_q} & x + k_q y \\ \downarrow \partial & & \downarrow \partial \\ (y, 0) & \xrightarrow{\psi_{q-1}} & y \end{array}$$

Thus, it suffices to show $\{Z_q \oplus Z_{q-1}\}$ has identity chain homotopic to 0. Indeed, take $D_q(x, y) = (0, x)$. Then

$$\begin{aligned} \partial_{q+1} D_q(x, y) + D_{q-1} \partial_q(x, y) &= \partial_{q+1}(0, x) + D_{q-1}(y, 0) \\ &= (x, 0) + (0, y) = (x, y). \end{aligned}$$

Claim: Let R be a PID and $f : C \rightarrow C'$ such that $H(Cf) = 0$. Then f is a chain homotopy equivalence.

Proof: Since $H(Cf) = 0$, we have $1_{Cf} \simeq 0$. Write $D_q : (Cf)_q \rightarrow (Cf)_{q+1}$

by $P_q = \pi_1 D_q i_1, Q_q = \pi_1 D_q i_2, R_q = \pi_2 D_q i_1, S_q = \pi_2 D_q i_2$. I claim that $D_q(x, y) = (P_q x + Q_q y, R_q x + S_q y)$. Indeed,

$$\begin{aligned} D_q(x, y) &= D_q(x, 0) + D_q(0, y) = D_q i_1(x) + D_q i_2(y) \\ &= (\pi_1 D_q i_1(x), \pi_2 D_q i_1(x)) + (\pi_1 D_q i_2(y), \pi_2 D_q i_2(y)) \\ &= (\pi_1 D_q i_1(x) + \pi_1 D_q i_2(y), \pi_2 D_q i_1(x) + \pi_2 D_q i_2(y)) \\ &= (P_q(x) + Q_q(y), R_q(x) + S_q(y)). \end{aligned}$$

Now, specializing $y = 0$,

$$\begin{aligned} (x, 0) &= \partial_{q+1}^f(P_q x, R_q x) + D_{q-1}(\partial_q x, 0) \\ &= (\partial_{q+1} P_q x + f_q R_q x, -\partial_q R_q x) + (P_{q-1} \partial_q x, R_{q-1} \partial_q x) \\ &= (\partial_{q+1} P_q x + f_q R_q x + P_{q-1} \partial_q x, R_{q-1} \partial_q x - \partial_q R_q x) \end{aligned}$$

so $x - f_q R_q x = \partial_{q+1} P_q x + P_{q-1} \partial_q x$ and R_q is a chain map. Now specialize $x = 0$:

$$\begin{aligned} (0, y) &= \partial_{q+1}^f(Q_q y, S_q y) + D_{q-1}(f_{q-1} y, -\partial_{q-1} y) \\ &= (\partial_{q+1} Q_q y + f_q S_q y, -\partial_q S_q y) + (P_{q-1} f_{q-1} y - Q_{q-1} \partial_{q-1} y, R_{q-1} f_{q-1} y - S_{q-1} \partial_{q-1} y) \end{aligned}$$

so $y - \partial_q S_q y + R_{q-1} f_{q-1} y - S_{q-1} \partial_{q-1} y \rightarrow y - R_{q-1} f_{q-1} y = \partial_q(-S_q)y + (-S_{q-1})\partial_{q-1}y$. Thus, f is a homotopy equivalence.

Claim: Let $\mathcal{V}$ be an open cover of X . Denote by $S(\mathcal{V})$ the set of simplices small of $\mathcal{V}$. Then the inclusion $S(\mathcal{V}) \rightarrow S(X)$ induces an isomorphism in homology.

Proof: Suppose $z \in Z_q(S(\mathcal{V}))$ such that $z = \partial_{q+1} w$ for some $w \in S_{q+1}(X)$. Then $\text{Sd}(z) = \text{Sd} \partial_{q+1}(z) = 0$. Note that $z - \text{Sd}(z) = \partial_{q+1} T_q(z) + T_{q-1} \partial_q(z) = \partial_{q+1} T_q(z)$. Since z is small of $\mathcal{V}$, so is $T_q(z)$. Thus $\partial_{q+1} T_q(z) \in B_q(S(\mathcal{V}))$.

Now suppose $z \in S_q(X)$. Pick $n \in \mathbb{Z}^+$ such that $\text{Sd}^n(z)$ is small of $\mathcal{V}$. We have $z - \text{Sd}(z) = \partial_{q+1} T_q(z)$, so $z \sim \text{Sd}(z)$ in $S_q(X)$. By induction, $z \sim \text{Sd}^n(z)$, and $\text{Sd}^n(z) \in Z_q(X) \cap S_q(\mathcal{V}) = Z_q(S(\mathcal{V}))$.

Claim: Let $\overline{U} \leq A^\circ$ (U need not be open and A need not be closed). Then $H_q(X - U, A - U) \rightarrow H_q(X, A)$ is an isomorphism.

Proof: Let $z \in Z_q(X - U, A - U)$. Suppose $z = \partial w + w'$ for some $w \in S_{q+1}(X)$ and $w' \in S_q(A)$. Then $\text{Sd}(z) = \text{Sd}(w')$. We have $z \sim \text{Sd}(z)$ in $H_q(X - U, A - U)$ and $\text{Sd}(z) \in S_q(X - U) \cap S_q(A) = S_q(A - U)$.

Now let $z \in Z_q(X, A)$. Pick $n \in \mathbb{Z}^+$ such that $\text{Sd}^n(z)$ is small of $\mathcal{V} = \{X - \overline{U}, A^\circ\}$. Let $\text{Sd}^n(z) = w + w'$ for $w \in S_q(X - U)$ and $w' \in S_q(A)$. Then $z \sim \text{Sd}^n(z)$ in $H_q(X, A)$ and $\text{Sd}^n(z) \sim w$ in $H_q(X, A)$, so $z \sim w$ in

$H_q(X, A)$. Write $z - w = \partial w_1 + w_2$ for $w_1 \in S_{q+1}(X)$ and $w_2 \in S_q(A)$. Then $\partial_q z - \partial_q w = \partial_q w_2$, so $\partial_q w = \partial_q z - \partial_q w_2 \in S_{q-1}(A) \cap S_{q-1}(X - U) = S_{q-1}(A - U)$. Thus, $w \in Z_q(X - U, A - U)$.

Suppose $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are cochain complexes such that $0 \rightarrow \mathcal{A} \xrightarrow{\mu} \mathcal{B} \xrightarrow{\varepsilon} \mathcal{C} \rightarrow 0$ is exact. Then $H^q(\mathcal{A}) \xrightarrow{H^q(\mu)} H^q(\mathcal{B}) \xrightarrow{H^q(\varepsilon)} H^q(\mathcal{C}) \xrightarrow{d_q} H^{q+1}(\mathcal{A}) \rightarrow \dots$ is exact.

Proof: First, suppose $a_q \in A_q$. Define $H_q(\mu)(\overline{a_q}) = \overline{\mu_q(a_q)}$. I claim that $H_q(\mu)$ is well-defined. Suppose $a_q - a'_q = \delta_{q-1}a_{q-1}$. Then $\mu_q a_q - \mu_q a'_q = \mu_q(a_q - a'_q) = \mu_q \delta_{q-1}a_{q-1} = \delta_{q-1}\mu_{q-1}a_{q-1} \in B^q(B)$. We can check that $H_q(\mu)$ is a module homomorphism. Suppose $H^q(\varepsilon)(\overline{b_q}) = 0$. Then $\varepsilon_q b_q = \delta_{q-1}c_{q-1}$. Pick b_{q-1} such that $\varepsilon_{q-1}b_{q-1} = c_{q-1}$. Then $\varepsilon_q b_q = \delta_{q-1}\varepsilon_{q-1}b_{q-1} = \varepsilon_q \delta_{q-1}b_{q-1}$, so $b_q - \delta_{q-1}b_{q-1} = \mu_q a_q$. Note that $0 = \delta_q b_q = \delta_q \mu_q a_q = \mu_{q+1} \delta_q a_q$, so $\delta_q a_q = 0$, ie $a_q \in Z^q(A)$. Thus, $H^q(\mu)(\overline{a_q}) = \overline{b_q}$. The others can be shown similarly.

Claim: $\text{Ext}(\mathbb{Q}, \mathbb{Z})$ is uncountable.

Proof: The sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ is exact, so we have the long exact sequence $\text{Hom}(\mathbb{Q}, \mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}) \rightarrow \text{Ext}(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \rightarrow \text{Ext}(\mathbb{Q}, \mathbb{Z}) \rightarrow \text{Ext}(\mathbb{Z}, \mathbb{Z})$. Note $\text{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$, $\text{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$, and $\text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0$. Hence, it suffices to show $\text{Ext}(\mathbb{Q}/\mathbb{Z}, \mathbb{Z})$ is uncountable.

Note that $\mathbb{Q}/\mathbb{Z}$ is Abelian with torsion equal to itself. Let A be such a group. I claim that $A = \bigoplus_{p \in \mathcal{P}} A_p$, where A_p denotes the p -torsion of A . Let $a \in A$ have order $n = \prod_{i=1}^k p_i^{e_i} > 0$. Let $n_i = \prod_{j \neq i} p_j^{e_j}$. Since $\gcd_{i \in [k]} n_i = 1$, pick x_i such that $\sum_{i=1}^k x_i n_i = 1$. Let $a_i = x_i n_i a$, so $a_i \in A_{p_i}$. Thus, we may write $a = \sum_{i=1}^k a_i$. Now suppose $0 = \sum_{i=1}^k a_i$. Pick e_i such that $p^{e_i} = |a_i|$ and let n_i be defined as before. Multiplying by n_i , we have $n_i a_i = 0$. Suppose $e_i > 0$. Then $p \mid p^{e_i} \mid n_i$, absurd. Thus, $e_i = 0$.

Let $A = \mathbb{Q}/\mathbb{Z}$ and A_p denote the p -torsion of $\mathbb{Q}/\mathbb{Z}$. We have $\text{Ext}(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) = \prod_{p \in \mathcal{P}} \text{Ext}(A_p, \mathbb{Z})$. It suffices to show $|\text{Ext}(A_p, \mathbb{Z})| \geq 2$ for all $p \in \mathcal{P}$. Suppose $\text{FTSOC } \text{Ext}(A_p, \mathbb{Z}) = 0$. Let $0 \rightarrow N_p \xrightarrow{j} F_p \rightarrow A_p \rightarrow 0$ be exact with N_p, F_p free. Since $\text{Ext}(A_p, \mathbb{Z}) = 0$, we assume j^* is surjective.

Since A_p has an element of order p , pick $x \in F_p$ such that $x \notin N_p$ and $px \in N_p$. Let $\{y_\alpha\}$ be a basis of N_p and write $px = \sum_{i=1}^k \lambda_i y_{\alpha_i}$. Since $x \notin N_p$, $p \nmid \lambda_{\alpha_i}$ for some i . Pick $\varphi : N_p \rightarrow \mathbb{Z}$ such that $\varphi(y_\alpha) = 0$ if $\alpha \neq \alpha_0$ and $\varphi(y_{\alpha_0}) = 1$ for the corresponding α_0 . Since j^* is surjective, pick $\tilde{\varphi} : F_p \rightarrow \mathbb{Z}$ such that $\tilde{\varphi}j = \varphi$. Then $\varphi(px) = \tilde{\varphi}(px) = p\tilde{\varphi}(x)$, absurd.

Claim: Let A be an abelian group, B an R -module, and R a commutative ring with identity. Then $\text{Hom}_R(R \otimes A, B) \cong \text{Hom}(A, B)$.

Proof: Let $\varphi : R \otimes A \rightarrow B$ be an R -homomorphism. Define $\Phi(\varphi)(a) = \varphi(1 \otimes a)$. I claim that $\Phi(\varphi)$ is indeed an abelian group homomorphism. Indeed,

$$\begin{aligned}\Phi(\varphi)(a + a') &= \varphi(1 \otimes (a + a')) = \varphi(1 \otimes a + 1 \otimes a') \\ &= \varphi(1 \otimes a) + \varphi(1 \otimes a') = \Phi(\varphi)(a) + \Phi(\varphi)(a')\end{aligned}$$

I claim that Φ is a group isomorphism. First,

$$\begin{aligned}\Phi(\varphi + \varphi')(a) &= (\varphi + \varphi')(1 \otimes a) = \varphi(1 \otimes a) + \varphi'(1 \otimes a) \\ &= \Phi(\varphi)(a) + \Phi(\varphi')(a),\end{aligned}$$

so Φ is a group homomorphism. Suppose $\Phi(\varphi) = 0$, so $\varphi(1 \otimes a) = 0$ for all a . Then $\varphi(r \otimes a) = r\varphi(1 \otimes a) = 0$. Since the elements $r \otimes a$ span $R \otimes A$, we have $\varphi = 0$. Conversely, suppose $\psi : A \rightarrow B$ is a group homomorphism. Define $f : R \times A \rightarrow B$ by $f(r, a) = r\psi(a)$. Note that

$$\begin{aligned}f(r + r', a) &= (r + r')\psi(a) = r\psi(a) + r'\psi(a) = f(r, a) + f(r', a) \\ f(r, a + a') &= r\psi(a + a') = r(\psi a + \psi a') = f(r, a) + f(r, a')\end{aligned}$$

so f is bilinear and descends to a map $\varphi : R \otimes A \rightarrow B$ such that $\varphi(r \otimes a) = r\psi(a)$. In particular, $\Phi(\varphi) = \psi$.